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CLINICAL RESEARCH ARTICLE

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Is there consensus on child violence exposure measures? A study of six recommended instruments

Ask Elklit [a], Morten Petterson [b], Amalie Salbøg Petersen [a] and Tine Nielsen [b]

[a]The Danish Centre of Psychotraumatology, University of Southern Denmark, Odense, Denmark; [b]Department of Applied Research in Education and Social Sciences, UCL University College, Odense, Denmark

ABSTRACT

Background: International research has established that children and adolescents are at high risk for being exposed to violence. A systematic review published in 2023 recommended six child and adolescent self-report violence against children (VAC) measures, based on their psychometric properties, in a systematic COnsensus-based Standards for the selection of health Measurement INstruments (COSMIN) review. However, the degree of overlap and heterogeneity was not part of the analysis.

Objective: To compare the six measures with respect to item overlap and differences concerning specification of exposure dimensions.

Method: A content analysis of the original 174 items resulted in a reduction to 38 unique items. These items were organized visually in a co-occurrence circle using an adapted version of Fried's R code. Furthermore, a pairwise comparison of event lists was performed using the Jaccard index.

Results: There was a modest overlap among the six measures. Only one item was present in all six measures, only two items were present in four measures, and 78% of the items were present in just one or two measures. The overall overlap between the six measures was 25%.

Conclusions: The lack of overlap among measures reflects a heterogeneity of definitions and purposes. It also impedes progress in research, as comparisons between various studies are difficult to make in a valid and reliable way. The lack of consensus also delays efficient political initiatives, because solid, consensual knowledge about the prevalence of VAC does not exist.

¿Existe consenso sobre las medidas de exposición a la violencia infantil? Un estudio de seis instrumentos recomendados

Antecedentes: La investigación internacional ha establecido que los niños y adolescentes corren un alto riesgo de estar expuestos a la violencia contra los niños (VAC en su sigla en inglés). Recientemente, en una revisión sistemática COSMIN (COnsensus-based Standards for the selection of health Measurement INstruments [Estándares basados en consenso para la selección de instrumentos de medida en salud]), se recomendaron seis medidas de exposición a la violencia hacia niños y adolescentes basadas en sus propiedades psicométricas. No obstante, el grado de solapamiento y heterogeneidad no formó parte del análisis.

Objetivo: Comparar las seis medidas con respecto al solapamiento de los ítems y las diferencias relativas a la especificación de las dimensiones de exposición.

Método: Un análisis de contenido de los 174 ítems originales dio como resultado una reducción a 38 ítems únicos. Estos ítems se organizaron visualmente en un círculo de co-ocurrencia utilizando una versión adaptada del código R de Fried. Además, se realizó una comparación por pares de las listas de eventos mediante el índice de Jaccard.

Resultados: Hubo un modesto solapamiento entre las seis medidas. Sólo un ítem estaba presente en las seis medidas; sólo dos ítems estaban presentes en cuatro medidas; el 78% de los ítems estaban presentes en sólo una o dos medidas. El solapamiento global entre las seis medidas fue del 25%.

Conclusiones: La falta de solapamiento entre las medidas refleja una heterogeneidad de definiciones y propósitos. Esto también obstaculiza el progreso de la investigación, ya que resulta difícil realizar comparaciones entre diversos estudios de forma válida y confiable. La falta de consenso también retrasa la adopción de iniciativas políticas eficaces, ya que no existe un conocimiento sólido y consensuado sobre la prevalencia.

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PALABRAS CLAVE
Violencia contra niños y adolescentes; instrumentos de exposición; ítems de escala; autoinforme; índice de Jaccard; círculo de acuerdo de Fried

Highlights

- Violence against children (VAC) is a high political and professional priority. A recent systematic review recommended six self-report measures based on their psychometric qualities.
- The VAC items were very diverse. A content analysis reduced the original 174 items to 38 unique items. A Jaccard index showed an overlap of 25%.
- The lack of consensus in definitions and applied measures impedes progress in research and delays important political, prophylactic initiatives.

CONTACT Ask Elklit aelklit@health.sdu.dk The Danish Centre of Psychotraumatology, University of Southern Denmark, Campusvej 55, DK-5320 Odense M, Denmark

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1. Introduction

Violence against children (VAC) is a major global issue that calls for action in prevention and remediation. Many studies on the prevalence, causes, and consequences of VAC have been published. However, the surge in studies has not been matched by gains in the quality of the metrics used (Meinck et al., 2023). Many instruments in the field of VAC have not been standardized. This contributes to huge variations in estimates of prevalence, which can hamper political initiatives and professional planning.

Good instruments are paramount for robust and precise estimation of the prevalence and impacts of VAC, and the effects of prevention and treatment intervention. Self-report measures of exposure to violence are important for the first-hand study of children aged 12 years or above, while parental reports are necessary with very young children. A systematic review by Meinck et al. (2023) identified all standardized current child and adolescent VAC self-report exposure measures and evaluated their psychometric properties to provide evidence-based guidance in selecting appropriate instruments for research.

This review followed the COnsensus-based Standards for the selection of health Measurement INstruments (COSMIN) guideline for systematic reviews of patient-reported outcome measures (Prinsen et al., 2018). The review by Meinck et al. (2023) resulted in the inclusion of 52 (39 original and 13 adaptations) measures from 124 studies that had published psychometric data. Most measures were not widely used, and some had been applied in many settings despite limited evidence for their psychometric qualities. Six instruments were recommended for use in research because of their sufficient psychometric qualities according to COSMIN guidelines: the Adverse Childhood Experiences (ACE), Childhood Trauma Questionnaire – Short Form (CTQ-SF), Parent–Child Conflict Tactics Scale (CTS-PC), Childhood Experiences of Care and Abuse Questionnaire (CECA.Q), ISPCAN Child Abuse Screening Tool (ICAST), and Juvenile Victimization Questionnaire (JVQ) (Figure 1).

The six instruments are quite diverse, and they are aimed at different purposes: population research, clinical assessment, and specific studies, such as examining changes in experiencing VAC during childhood and adolescence (see Section 2). The overall purpose, the target population, and the sociocultural context are important factors when a self-reported instrument is selected. The diversity in the six instruments raises several questions: What is violence? What types of violence exist? When specific behaviours are grouped into one type/class, what are the relationships within the class and between the classes? When a push is classified as a physical violence item, how does it relate to harsher violence, such as being burnt, stabbed, or shot? How does the instrument distinguish between mild and serious acts within the same category? Can a push also be a sign of friendship, intimate contact, or playful behaviour?

The variety of scales can be understood as ways to flexibly meet diverse challenges and interests in research and in clinical practice. The variety means, however, that the accumulation of research knowledge could be impeded if the scales are different and measure a large selection of events that may not be comparable. This potential danger could be one reason why studies on the prevalence of child maltreatment vary enormously, and this fact frustrates efforts in this important research field to develop trustworthy results that can inform political initiatives to prevent and mitigate the consequences of VAC.

Dohrenwend and Dohrenwend (1974) pointed out that decisions made in the construction of an event list will ultimately determine the kinds of inferences and generalizations that can be made: How do we define the events to be sampled and what is the population of events from which the sample is to be drawn? As the basic types of VAC all have physical, psychological, and social components, it becomes paramount to have clear definitions and try to reach consensus on the common ground. With this in mind, a close analysis of the six instruments selected as being of the best quality by Meinck et al. (2023) could be a starting point for identifying the elements of the common ground and understanding the diversity embedded in operationalization of the potentially traumatic acts of VAC.

The aim of the present study is, therefore, to study the overlap of items among the six most widely used and psychometrically sound instruments recording VAC events identified by Meinck et al. (2023). Furthermore, do these six instruments have a common core? If that is the case, we may, in the future, be able to pool the items representing this core and thereby accumulate more valid and useful knowledge on the prevalence of VAC.If it is not the case, this could a starting point to reflect on how to progress the development of self-report instruments with sufficient content validity in the field of VAC.

2. Methods

The six instruments assessed in this study are described as follows.

- *Adverse Childhood Experiences (ACE)*. Six versions of the ACE are mentioned in the review by Meinck et al. (2023): (1) ACE, used in Japan; (2) ACE-IQ, used in Malawi; (3) ACE-ASF, used in South Africa; (4) M-ACE, used in the USA; (5) ACE-BRFSS; and (6) ACE-S. There is no further information on the last two of these (regarding authors and publications in Supplements 4 and 8 in the COSMIN study). The number of items varies from nine in

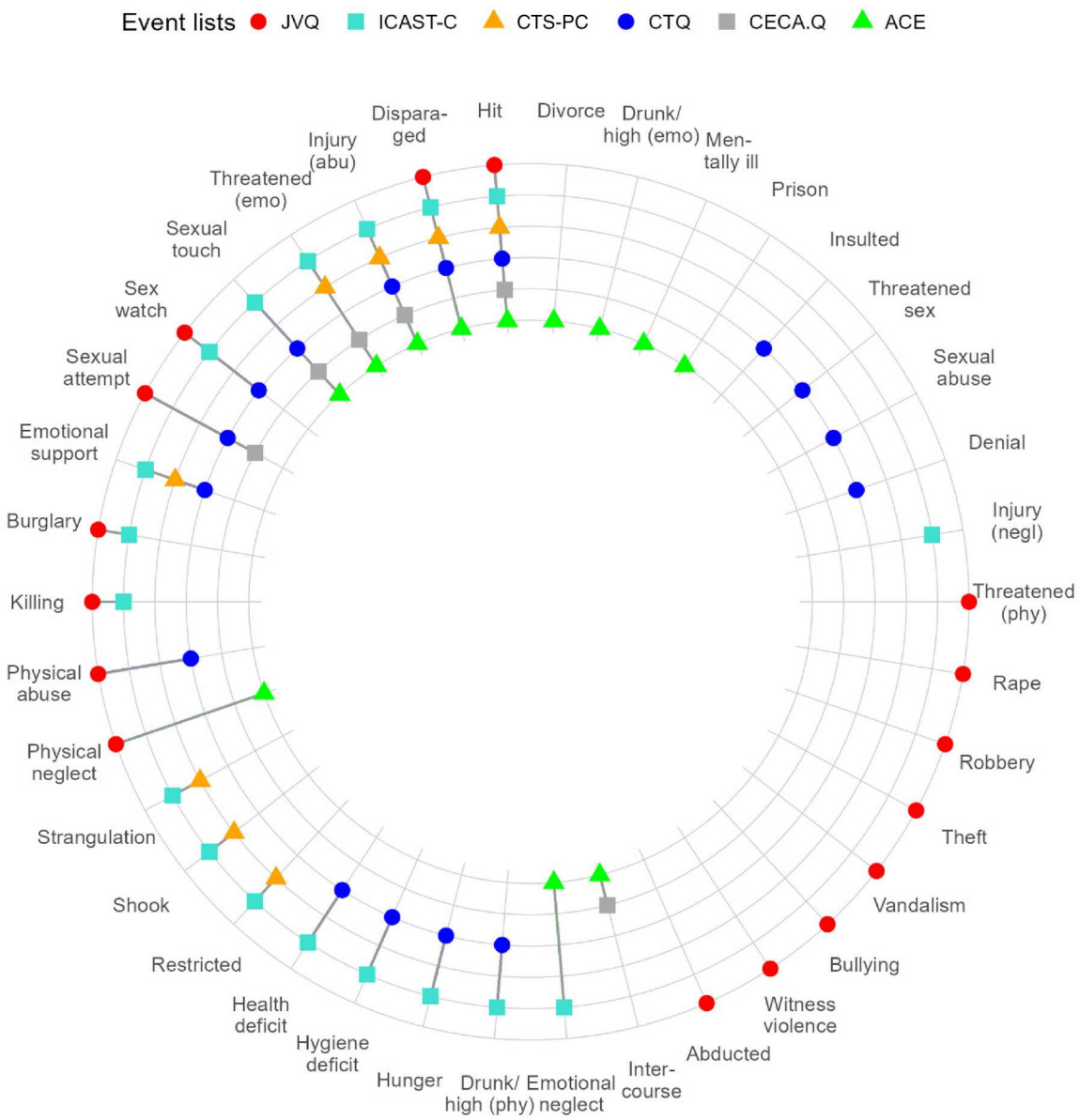


Figure 1. Overlap of 38 pure events in six traumatic event lists. JVQ = Juvenile Victimization Questionnaire; ICAST-C = ISPCAN Child Abuse Screening Tool – Child version; CTS-PC = Parent–Child Conflict Tactics Scale; CTQ = Childhood Trauma Questionnaire; CECA.Q = Childhood Experiences of Care and Abuse Questionnaire; ACE = Adverse Childhood Experiences.

the ACE version (not 14 as stated in Table 1 in the COSMIN study) to 11 and 14 in the other versions, but there is no mention of the 17 items used in the first publication (Felitti et al., 1998). The ACE-BRFSS version was changed in 2021 from 11 to 13 items. To avoid more confusion, we have therefore chosen to use the often-mentioned 10-item version used in Finkelhor et al. (2013). Meinck et al. (2023) report in their Table 1 that a five-point Likert scale is used; however, this is not correct: it is a yes/no format. The questionnaire assesses five types of childhood trauma measured in the ACE study, which are personal: physical abuse, verbal abuse, sexual abuse, physical neglect, and emotional neglect. The other five concern family members and their behaviour. They include, for example, parental mental illness and divorce/separation. They are not considered traumatic as per criterion A of the Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5), but still count as adverse childhood experiences because of their association with negative outcomes.

- *Childhood Experiences of Care and Abuse Questionnaire (CECA.Q).* This questionnaire was originally used to investigate lifetime risk factors for psychological disorders (Bifulco et al., 2005). The questionnaire was sent to participants' homes, and assessed lack of parental care (neglect and antipathy), parental physical abuse, and sexual abuse by any adult before the age of 17 years. It has introductory screening questions for sexual and physical abuse. They are followed by 10 items with a yes/no format, with three items related to sexual abuse where the respondent also has the option to answer 'unsure'. In addition, there are questions about age at the first experience,

frequency, and relationship to the perpetrator(s). The scale for parental care is divided into father and mother scales with 16 items; responses to these items are given on a five-point scale from 1 ('yes, definitely) to 5 ('no, not at all'), with a midpoint value of 3 ('unsure'). Two scales, called 'antipathy' and 'neglect', seem not to be a part of the COSMIN analysis, and for this reason, we did not include them either.

- *Childhood Trauma Questionnaire – Short Form (CTQ-SF)*. The CTQ-SF (Bernstein et al., 2003) contains 28 items covering five dimensions: physical and emotional abuse, emotional neglect, sexual abuse, and physical neglect, plus three validity items. Response options vary from 1 ('never true)' to 5 ('very often true').
- *Parent–Child Conflict Tactics Scale (CTS-PC)*. The CTS-PC (Straus et al., 1998) is a multi-item measure developed to assess parent-to-child conflict, based on the Conflict Tactics Scale (CTS1). It includes 22 items on non-violent discipline, psychological aggression, physical assault, and neglect. Response options are rated on an eight-point scale from 'never' to 'more than 20 times'.
- *ISPCAN Child Abuse Screening Tool*. The International Society for Prevention of Child Abuse and Neglect (ISPCAN) Child Abuse Screening Tool exists in several versions. In the COSMIN study, the authors use the Retrospective version (ICAST-R) (Dunne et al., 2009), which was developed for young adults, and the not the Child version (ICAST-C) (Zolotor et al., 2009), for use in children and adolescents. Both were developed to survey the prevalence of childhood victimization. In addition to demographic questions, the 61 items serve as screeners, and positive replies are followed by queries on frequency (eight possibilities, from 'no answer' and 'never in my life' to ≥ 50 times) and perpetrator (response format: yes/no to four possibilities: adult male, adult female, child/adolescent male, child/adolescent female).
- *Juvenile Victimization Questionnaire (JVQ)*. The JVQ (Finkelhor et al., 2005) is designed to provide a quantified description of all the major forms of offences against youth, including conventional crime, child maltreatment, peer and sibling victimization, sexual victimization, and witnessing and indirect victimization. It has 34 items, to be answered by yes/no.

3. Analysis

3.1. Content analysis

The first analysis conducted used the 174 events listed in the six VAC exposure instruments (see Section 2). After organizing the 174 events, these were analysed by the first and last authors to achieve a list of 'pure' events. This process involved a lot of reflection and choices. If one starts from an understanding of VAC as intentional acts resulting in pain or injury, a number of items are not covered by this definition, because either they were descriptions of inner states ('shock, upset') or experienced emotions in other people ('hatred'), or they were social events ('divorce, prison'), or they were something happening in another person ('lost control, drunk'). Another type of problem was most pronounced in the ACE scale, which has a few items, but these items are overloaded, with up to five different acts within one item (from being 'sexually touched' to 'intercourse'; from being 'pushed' to 'being hit so hard that you were injured'). A third type of problem involved situations where the act might not have been intended to harm, might not be directed towards the child, and might not have resulted in pain or injury, e.g. 'forbade you from going out, time-out, took away privileges or money'. A fourth problem was the inclusion of emotional support, which is only a VAC act if it is totally absent, and then it is categorized as emotional neglect or emotional abuse. A final problem related to items that were general statements intended to reveal denial, e.g. 'I had a perfect childhood'.

The all-inclusive list of 174 items was eventually reduced to 38 'pure' or unique events (Table 1 and Figure 1), where information about the gender and the status/role of the perpetrator was excluded.

As is evident from Table 1, some of the previously mentioned problems persist in this list. Thus, in three cases, we have kept the option of scoring acts as examples of different types of VAC: an injury can be due to neglect ('my dad didn't fix my bike, so I had a crash and broke my hand') or due to abuse; parents being drunk or high can be associated only with emotional neglect or with physical neglect (or both); threats can be emotional but they can also have such a violent character that most people would conceive them as physical abuse because they are so close to physical aggression.

The pure event list was then further discussed and resulted in a list of 13 event types (see Table 1 and Figure 3), consisting of both the well-established child maltreatment types and additional event types (physical abuse, emotional abuse, sexual abuse, physical neglect, emotional support, burglary, loss, emotional neglect, witnessing, vandalism, theft, robbery, and denial) that may be considered potential traumatic events and acts or processes associated with resilience or defense mechanisms. These two lists of pure events and event types form the basis for the following analyses. The large table containing information on the 174 original events and the relationship to the pure events and event types is shown in the Supplemental file.

Table 1. The 38 pure events divided into 13 event types.

	Pure event		Event type
1	Disparaged	1	Emotional abuse
2	Threatened emotionally	1	Emotional abuse
36	Threatened physically	3	Physical abuse
3	Injury by neglect	2	Physical neglect
37	Injury by abuse	3	Physical abuse
4	Sexual touch	4	Sexual abuse
5	Intercourse	4	Sexual abuse
6	Emotional neglect	5	Emotional neglect
7	Physical neglect	2	Physical neglect
8	Divorce	5	Emotional neglect
9	Drunk/high emotional	5	Emotional neglect
38	Drunk/high physical	2	Physical neglect
10	Hit	3	Physical abuse
11	Mentally ill	5	Emotional neglect
12	Prison	5	Emotional neglect
13	Insulted	1	Emotional abuse
14	Physical abuse	3	Physical abuse
15	Threatened sex	4	Sexual abuse
16	Sexual attempt	4	Sexual abuse
17	Sex watch	4	Sexual abuse
18	Sexual abuse	4	Sexual abuse
19	Emotional support	6	Emotional support
20	Hunger	2	Physical neglect
21	Hygiene deficit	2	Physical neglect
22	Health deficit	2	Physical neglect
23	Denial	12	Denial
24	Restricted	3	Physical abuse
25	Shook	3	Physical abuse
26	Strangulation	3	Physical abuse
27	Killing	7	Loss
28	Burglary	13	Burglary
29	Rape	4	Sexual abuse
30	Robbery	9	Robbery
31	Theft	10	Theft
32	Vandalism	11	Vandalism
33	Bullying	1	Emotional abuse
34	Witnessed violence	8	Witnessing
35	Abducted	3	Physical abuse

The number of pure events that appeared across the number of event lists was calculated, and a matrix of pure events and traumatic event lists was generated, so that a pure event was given a value of 1 if it appeared in an event list, and a value of 0 if it did not. The data matrix is publicly available at Zenodo.org (https://zenodo.org/doi/10.5281/zenodo.10124102). The matrix also contains a column informing on the type of event under which each pure event was categorized.

3.2. Overlap in pure events across traumatic event lists

The pure event by event type matrix resulting from the content analysis was used to produce a visual representation of the overlap of pure events across the event lists in a co-occurrence circle, using an adapted-for-purpose version of the R code provided by Fried (2016, 2020) (see Section 3.3 for an account of the adaptations), as well as a numerical evaluation across the entire event lists, using the Jaccard index (Jaccard, 1912). Fried's (2016) co-occurrence circle in combination with the Jaccard index has gained usage in evaluating overlap in various psychological symptoms scales since Fried's first use with depression scales, e.g. across youth obsessive–compulsive disorder scales (Visontay et al., 2019), anxiety scales (Wall & Lee, 2021), and, not least, adult traumatic event lists (Karstoft & Armour, 2022).

The Jaccard index is a similarity index using a data matrix, with binary data coded 1 for the presence of an event in a list and 0 for no presence of the event. The Jaccard index is calculated by $s/(u_1 + u_2 + s)$, where s is the number of pure events included in two event lists, and u_1 and u_2 are the numbers of pure events that are unique to each of the two event lists. This was calculated for all pairs of event lists. The Jaccard index takes values from 0 (no overlap across lists) to 1 (complete overlap across lists).

3.3. The adapted R code

The method used for constructing the co-occurrence circle followed the approach outlined in Fried (2016, 2020), with two key modifications to enhance the functionality of the code. First, the data manipulation process, originally executed using the data.table package (Dowle & Srinivasan, 2023), was transitioned to the tidyverse package (version 2.0.0) (Wickham et al., 2019). This transition not only ensures enhanced clarity and readability of the code but also makes it more accessible to others, as the tidyverse ecosystem is well documented and supported by a large, active community. Secondly, we introduced a new approach to determine the order of items – in this context events – along the perimeter of the circle to increase visual informativeness. Specifically, events were arranged anti-clockwise based on their frequency, starting with those events that were most prevalent across event lists and concluding with those found in the fewest event lists. In cases where events shared equal frequency of co-occurrence, we applied a sorting criterion based on their presence in the outermost circle and moving inwards. In this way, we would first plot the event(s) with the highest frequency of co-occurrence, then the event(s) with the second highest co-occurrence, starting with the event occurring most towards the outer rim of the co-occurrence circle, and so forth. In this manner, we achieved a co-occurrence circle with an inward-moving plotting of events, and we found that this provided a clearer and more insightful representation of the relationship between events and event lists, thus enhancing the overall interpretability of the co-occurrence circle. The R code for both the agreement circles and the Jaccard index is publicly available at Zenodo.org (https://zenodo.org/doi/10.5281/zenodo.10124040).

4. Results

The content analysis resulted in a pool of 38 pure events falling under 13 different event types (Table 1). Of the 38 pure events, the majority (78.9%) appeared in only one or two event lists (Table 2).

Table 2. Number of pure events that appeared across different numbers of event lists.

Number of pure events	In number of event lists	Percentage of pure events
17	1	44.7%
13	2	34.2%
3	3	8.9%
2	4	5.3%
2	5	5.3%
1	6	2.6%

Table 3. Number of pure events and event types in the six event lists.

Event lists	Number of pure events	Number of event types covered
JVQ	16	10
ICAST-C	18	8
CTS-PC	8	3
CTQ-SF	16	6
CECA.Q	6	3
ACE	12	5

Note: JVQ = Juvenile Victimization Questionnaire; ICAST-C = ISPCAN Child Abuse Screening Tool – Child version; CTS-PC = Parent–Child Conflict Tactics Scale; CTQ-SF = Childhood Trauma Questionnaire – Short Form; CECA.Q = Childhood Experiences of Care and Abuse Questionnaire; ACE = Adverse Childhood Experiences.

Two pure events (i.e. 'threatened emotionally' and 'sexual touch') appeared in four event lists, but not the same lists. Both of these pure events appeared in ICAST-C, CECA.Q, and ACE, while 'threatened emotionally' also appeared in CTS-C and 'sexual touch' additionally in CTQ-SF. A further two pure events (i.e. 'injury by abuse' and 'disparaged') appeared in five event lists; however, each was lacking from a different list: 'injury by abuse' did not appear in JVQ and 'disparaged' did not appear in CECA.Q. Only a single pure event (i.e. 'hit') was represented in all six traumatic event lists (Table 2 and Figure 1).

ICAST-C had the highest number of pure events represented in its items (18), both JVQ and CTQ-SF had almost as many pure events represented (16), while CECA.Q had the fewest pure events represented (six) (Table 3). No single event list contained items covering all 13 event types. The highest number of event types covered was 10 (JVQ) and the lowest was three (CTS-PC and CECA.Q). See Figures 2 and 3 for an illustration of the pure events as they relate to event types across the six instruments and how the specific event types are represented in instruments.

Turning to the overlap across the event lists, we found that there was the greatest similarity between ICAST-S and CTS-PC (44.4%), ICAST-C and CTQ-SF (41.7%), and CECA.Q and ACE (38.5%) (Table 4). The least similar event lists in terms of overlap were JVQ and CTS-PC (9.1%), and JVQ and CECA.Q (10.0%).

5. Discussion and implications

When researchers want to study VAC, to find out about the prevalence and sequelae of VAC, it is sensible to use the same definitions and terms. The collection of studied measures in Meinck et al. (2023) reveals a plethora of different items building on different juridical, psychological, and sociological definitions to cover the field of VAC, ranging from epidemiological screening to clinical assessment. Some of the recommended measures have been developed for young adults and not for use with children or adolescents. Ideally, all measures should be tailored to be developmentally sensitive, recognizing that children and adolescents face different challenges as they age. In addition, considering that some may be particularly vulnerable as a result of exposure to violence, the way in which the measure is introduced should also be sensitive and provide opportunities for assistance. It is noteworthy that only one measure seems to have been developed to suit young children (4–8 years old): a picture-based, modularized child report version of the Parent–Child Conflict Tactics Scale – Revised (Sierau et al., 2018). Meinck et al. (2023) also stress that all measures should be culturally adapted and tested before use to see whether they are suitable for the specific population.

The aims of the present study were to investigate the congruence or overlap of six recommended instruments with the best psychometric qualities which came out of a COSMIN study (Meinck et al., 2023). The overall finding is that this overlap is quite modest; the highest overlap in events between two measures was 44%, while the lowest was just 8%, and the average for all six was 25%. This finding of modest overlap is similar to the main finding of Karstoft and Armour (2022), who studied eight adult trauma exposure measures. Such a modest overlap between measures that intend to cover the same field constitutes a challenge or a threat to the accumulation of solid and accepted general knowledge concerning VAC. However, there is full agreement that VAC includes physical and emotional abuse, and partial agreement that sexual (five out of six lists) and physical neglect (four out of six lists) should be included. This could suggest a consensus on a common core of acts that constitute VAC, although one might argue that other acts could also be included; for example, in some cases, physical neglect is not due to intentions to harm but to lack of parental ability, and in these cases the child has no experience of violence from the parent. A clinical example is a child who grows up with an intellectually disabled, single mother who cannot cook a meal or wash clothes. This case is an example of child maltreatment, but not of VAC.

One might also argue that events other than violence could be harmful and destructive to children, and therefore deserve attention and should be included in the 'common core'. One such candidate

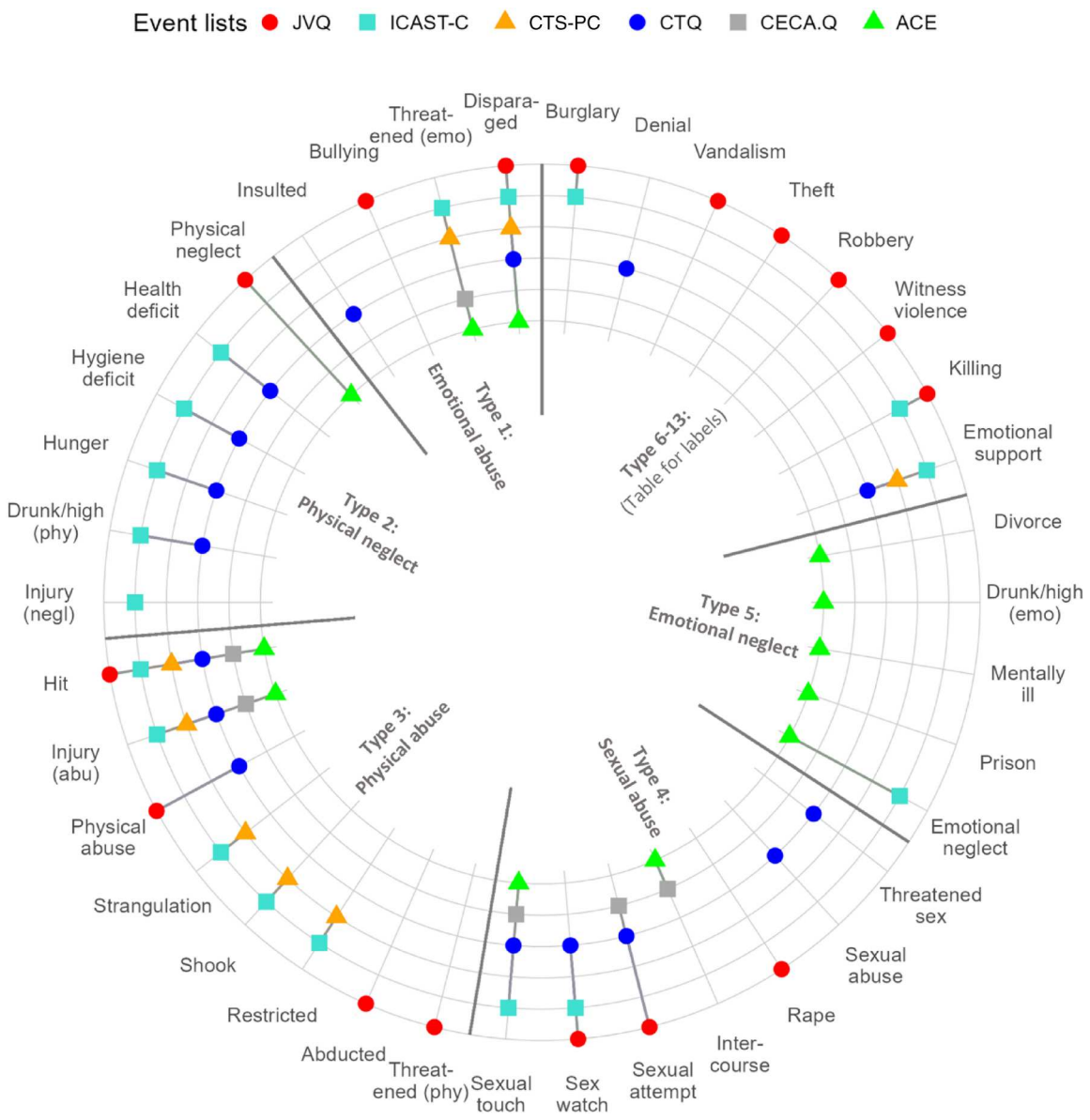


Figure 2. Overlap of 38 pure events within types of events in six traumatic event lists. (Types 6–13: see Table 1 for labels.) JVQ = Juvenile Victimization Questionnaire; ICAST-C = ISPCAN Child Abuse Screening Tool – Child version; CTS-PC = Parent–Child Conflict Tactics Scale; CTQ = Childhood Trauma Questionnaire; CECA.Q = Childhood Experiences of Care and Abuse Questionnaire; ACE = Adverse Childhood Experiences.

could be witnessing domestic violence, e.g. parents fighting or a parent who physically punishes a sibling, which is well evidenced as being as harmful to children as being physically abused themselves (Kitzmann et al., 2003).

If the purpose of the investigation is to encompass other types of aversive life events or traumatic exposure, this could be done in separate measures or as a separate part of VAC instruments. One good argument for that is the common finding of accumulation of child maltreatment, traumas, and adverse life events (Elklit & Murphy, 2022; Felitti et al., 1998; Shevlin & Elkli, 2008).

Our recommendation to advance the field is to focus on the overall interest, which is to estimate the extent of VAC and the sequelae of VAC. Two approaches stand out: the event-centred and the person-centred approaches. Two targets are common: a broad approach and a narrower approach. It is legitimate to have a broad approach to find out the incidence of every push and every unwanted sexual or derogatory remark. The flipside to this approach is that there are many items with few or no empirical personal consequences, and it may lead to passivity when it comes to prophylactics because VAC is common, widespread, a universal problem, and so extensive that it is untreatable. A narrower target using a person-centred approach, with latent class analysis covering the four main types of VAC, could be useful to inform practitioners and politicians, identifying the precise extent of sexual, psychological, and physical abuse that needs to be considered for intervention. By combining the two approaches – first the event and then the

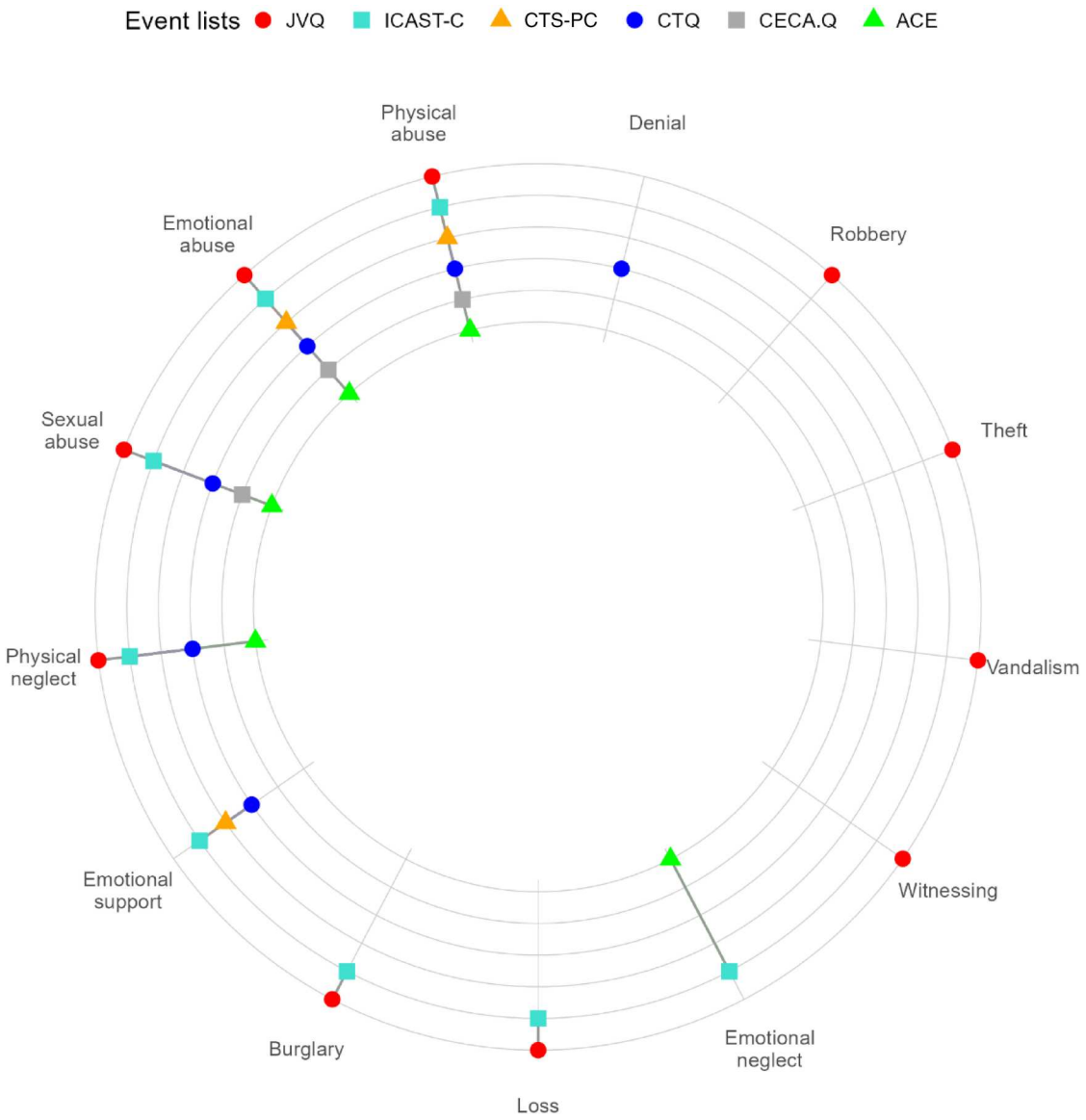


Figure 3. Overlap of 13 traumatic event types across six event lists. JVQ = Juvenile Victimization Questionnaire; ICAST-C = ISPCAN Child Abuse Screening Tool – Child version; CTS-PC = Parent–Child Conflict Tactics Scale; CTQ = Childhood Trauma Questionnaire; CECA.Q = Childhood Experiences of Care and Abuse Questionnaire; ACE = Adverse Childhood Experiences.

person – it will be possible to conduct comparative and cross-cultural VAC studies where the results are not arbitrary comparisons that depend totally on the applied instruments.

Table 4. Jaccard similarity indices showing overlap in pure events between event lists.

	JVQ	ICAST-C	CTS-PC	CTQ-SF	CECA.Q	ACE
JVQ		0.172	0.091†	0.185	0.100†	0.120
ICAST-C	0.172		0.444*	0.417*	0.200	0.250
CTS-PC	0.091†	0.444*		0.200	0.273	0.250
CTQ-SF	0.185	0.417*	0.200		0.222	0.167
CECA.Q	0.100†	0.200	0.273	0.222		0.385*
ACE	0.120	0.250	0.250	0.167	0.385*	
Mean overlap	0.080†	0.206	0.162	0.165	0.164	0.163

Note: The Jaccard similarity index ranges from 0 (no overlap) to 1 (total overlap). *Overlaps of 30% or more; †overlaps of 10% or less.
JVQ = Juvenile Victimization Questionnaire; ICAST-C = ISPCAN Child Abuse Screening Tool – Child version; CTS-PC = Parent–Child Conflict Tactics Scale; CTQ-SF = Childhood Trauma Questionnaire – Short Form; CECA.Q = Childhood Experiences of Care and Abuse Questionnaire; ACE = Adverse Childhood Experiences.

6. Conclusions

VAC is a political and professional priority. A systematic review recommended six measures of child and adolescent self-report measures to be used based on their psychometric qualities. We investigated the definitions and characteristics of all of these items, and scrutinized them in a content analysis and thereby reduced them to 38 'pure' items. We then studied the overlap among the measures, which was as modest as 25%. Some of the recommended included measures appeared to be used in older age groups. There was only one measure for young children, aged 4–8 years, that was developmentally sensitive, a picture-based version; however, this was not among those that were recommended.

The review by Meinck et al. (2023) recommends that measures should be culturally adapted and tested before use to see whether they are fit for a specific

population. This is certainly warranted; however, there is strong evidence that this procedure is seldom followed. The heterogeneity in definitions and measures impedes progress in research, particularly in national and cross-national comparisons of prevalence rates of VAC. It also thwarts studies on the consequences of VAC and studies on intermediate factors, because the starting points and the measures of VAC differ from study to study. This eventually has a negative effect on political prophylactic initiatives because it is unlikely that any consensus will exist concerning the prevalence of VAC.

Disclosure statement

No potential conflict of interest was reported by the authors.

Ethical approval

Ethical approval was not applicable for this secondary data analysis.

Data availability

Data will be made available upon reasonable request to the corresponding author.

ORCID

Ask Elklit http://orcid.org/0000-0002-8469-7372
Morten Petterson http://orcid.org/0009-0005-7102-8144
Tine Nielsen http://orcid.org/0000-0002-9087-729X

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